Module 06 · Lecture 5

# The cognitive walkthrough

Four questions about a sketch, asked at every step of a mission

**Álvaro Figueira**
Interação Pessoa-Máquina · DCC · FCUP



1

## Summary

- Why evaluate a sketch at all, and what a cognitive walkthrough is
- One example for the whole module: an operator, a mission at Touvedo, a sketch
- The four questions, asked one step at a time, and the table they produce
- Phase 1 of the project: the same table, on another group's sketches



2

# Why evaluate a sketch at all

**Phase 1**

**Cognitive walkthrough on the sketches**

This module. Session S6, November.



**Fixing a problem: a redraw, minutes**

**Phase 2**

**Heuristic evaluation on the wireframes**

Module 10. Session S8.



**Fixing a problem: an hour in Figma**

**Phase 3**

**Usability tests on the prototype**

Module 11. Session S9, five external testers.



**Fixing a problem: a day**

**Inspection methods find 30 to 90 % of the problems that user tests find** (Hollingsed and Novick, 2007). One evaluator alone finds about a third of them, so Phase 1 uses two or three evaluators per sketch. No lab, no recruiting: hours, not a week (Nielsen, 1989).

**Question for the class.** Which problems can you find on a paper sketch, and which ones only appear with real users?

Nielsen, Usability engineering at a discount, 1989. Hollingsed and Novick, SIGDOC 2007



3

# What a cognitive walkthrough is

An evaluator who did not design the interface **simulates a first-time user doing one task, one step at a time, on a sketch**, and asks at each step: would this user know what to do, and would they understand what happened?

### Who does it

People with distance from the design: another group, a colleague, an analyst from outside.

**The more distance, the more they see**. They do not know what the designer "meant".

**No experts needed**. Two or three evaluators per sketch.

### What it finds

Confusion, dead ends, missing feedback, wrong labels, steps nobody drew.

Writing the correct action sequence exposes screens nobody sketched.

Found step by step, instead of by impression.

### What it does not find

Speed and the errors of trained users. Those need real users (Module 11).

Only the tasks you chose. Choose the three missions that matter.

Each evaluator finds part of the problems, with some false alarms.

**Module 05:** the method **evaluates the exploratory stage**, where every one of the seven stages is conscious.
The first week at the COL is a good situation for this method.

Lewis, Polson, Wharton and Rieman, CHI 1990. Wharton, Rieman, Lewis and Polson, 1994



4

# The case for this module: one operator, one mission, one sketch

**Tiago, a COL operator**

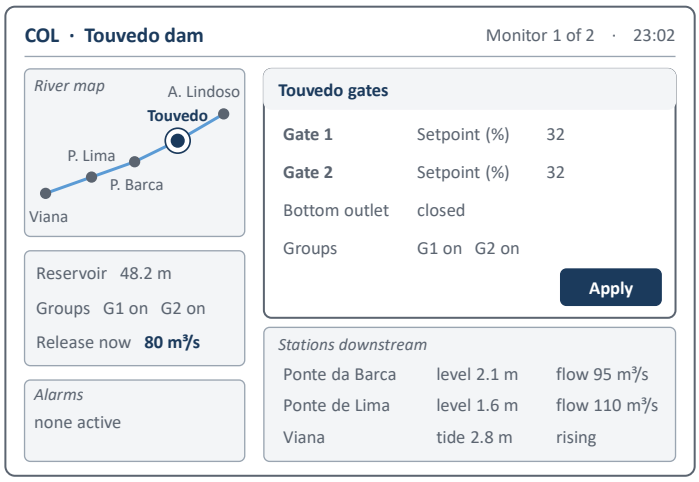
First week on the new interface. Trained on the old system: he knows the river and the gates, not these screens.

## Mission

**"Raise the release at Touvedo from 80 to 120 m³/s"**

The correct action sequence, written by the design group (the mission of Module 05, slide 19):

1. Select Touvedo on the river map
2. Find the gate controls on the panel
3. Enter 120 in the flow field
4. Check Ponte da Barca before confirming
5. Confirm the order




Hypothetical sketch of a COL gate panel, drawn for this module. Yours will fail in other places, and that is what Phase 1 is for.



5

# The four questions* and a failure for each one

*Wharton, Rieman, Lewis e Polson (1994)

**Q1 Will the user try to achieve the right effect?**

Stages 1 and 2: goal and intention

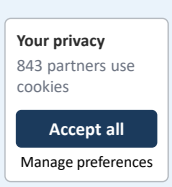
The start button that does nothing until the brake is pressed. Nobody knows that step exists, so nobody tries it.




**Q2 Will the user notice that the correct action is available?**

Stage 3: the sequence. Visibility

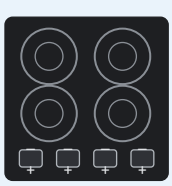
"Reject all" hidden behind "Manage preferences". The action exists, and nobody sees it.




**Q3 Will the user associate the correct action with the effect they want?**

Stages 3 and 4: labels and mapping

Four rings, four buttons in a row. Which button heats which ring?



**Q4 If the correct action is performed, will the user see progress being made?**

Stages 5 to 7: feedback and evaluation

The lift button that does not light up. Everyone presses it three times.




**Spencer (CHI 2000), two questions when time is short:** will the user **know what to do** at this step? If they do the right thing, **will they know they did**, and that they are making progress?

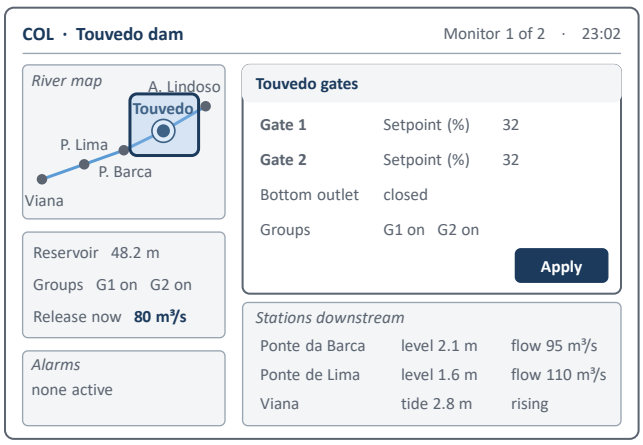


6

# Select Touvedo on the river map

STEP 1 OF 5

**Tiago, COL operator, first week on the new interface.** **Mission:** raise the release at Touvedo from 80 to 120 m³/s.




**Q1** yes — **Knows this step is needed?**
The mission names Touvedo, and Tiago knows the release is changed there.

**Q2** yes — **Sees that the action is available?**
The map is the first thing on the screen, and Touvedo is on it.

**Q3** yes — **Connects the action to the effect?**
A dam on a map: touching it opens the dam. The old system worked the same way.

**Q4** yes — **Sees progress after acting?**
Touvedo is highlighted and the header reads "Touvedo dam".

**No problem at this step.** A success needs a reason too: the map, the highlight and the header are what make Tiago do the right thing.

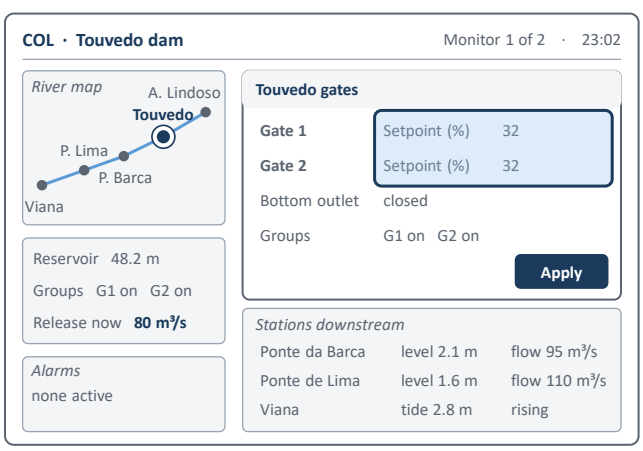


7

# Find the gate controls on the panel

STEP 2 OF 5

**Tiago, COL operator, first week on the new interface.** **Mission:** raise the release at Touvedo from 80 to 120 m³/s.




**Q1** yes — **Knows this step is needed?**
Tiago knows the release is changed at the gates.

**Q2** no — **Sees that the action is available?**
The setpoints are plain numbers, in the same style as the read-only values around them. Nothing shows they can be edited.

**Q3** yes — **Connects the action to the effect?**
Once he touches one, "Setpoint" says what it is.

**Q4** yes — **Sees progress after acting?**
A cursor appears and the number can be changed.

**Problem found (Q2).** A control that looks like a display. Tiago reads the panel and sees nothing to act on. Severity 3, major.

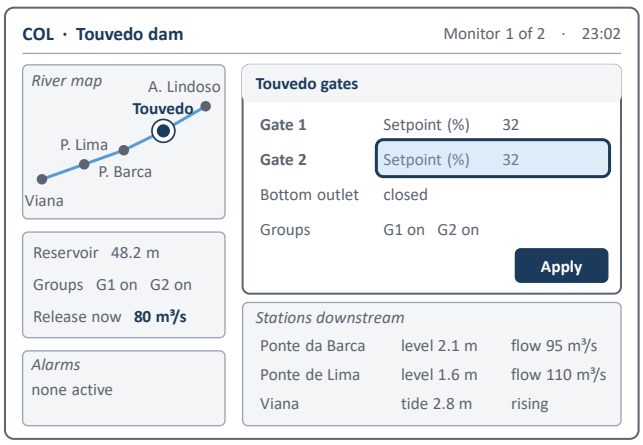


8

# Enter 120 in the flow field

STEP 3 OF 5

**Tiago, COL operator, first week on the new interface.** **Mission:** raise the release at Touvedo from 80 to 120 m³/s.




**Q1** yes — **Knows this step is needed?**
Tiago wants 120 m³/s and looks for where to type it.

**Q2** yes — **Sees that the action is available?**
There is one editable field in the panel.

**Q3** no — **Connects the action to the effect?**
The field reads "Setpoint (%)": gate opening, in percent. Tiago wants m³/s and does not know the opening-to-flow curve.

**Q4** yes — **Sees progress after acting?**
The typed value appears in the field (but see step 5).

**Problem found (Q3).** A mechanism term where Tiago expects a task term. He may type 120 %, which opens the gate fully, or give up. Severity 4, blocks the task.

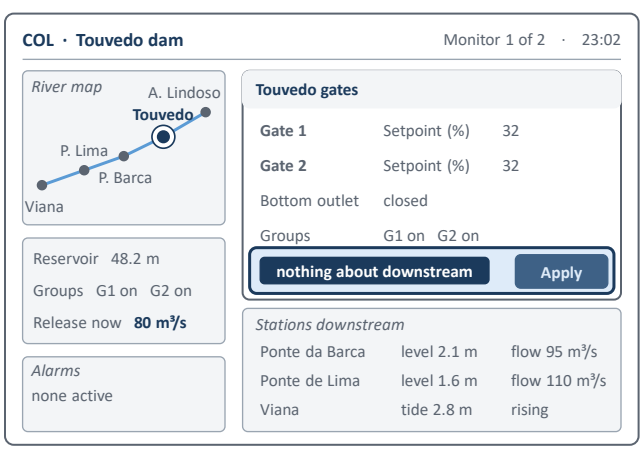


9

# Check Ponte da Barca before confirming

STEP 4 OF 5

**Tiago, COL operator, first week on the new interface.** **Mission:** raise the release at Touvedo from 80 to 120 m³/s.




**Q1** no — **Knows this step is needed?**
Nothing on the panel says to check downstream before confirming. In the old system, a colleague phoned Ponte da Barca.

**Q2** yes — **Sees that the action is available?**
The Stations panel is on the screen, under the gates.

**Q3** yes — **Connects the action to the effect?**
Ponte da Barca is the first station listed.

**Q4** yes — **Sees progress after acting?**
The panel shows its level and flow.

**Problem found (Q1).** The rule "check downstream before raising the release" lives in people's heads, not on the screen. Tiago never forms the goal. Severity 3, major.

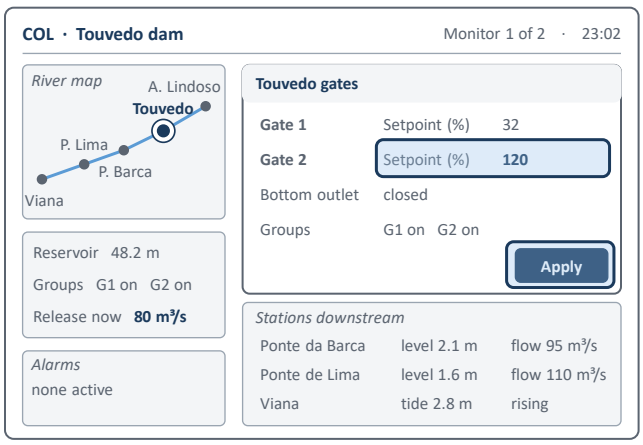


10

# Confirm the order

STEP 5 OF 5

**Tiago, COL operator, first week on the new interface.** **Mission:** raise the release at Touvedo from 80 to 120 m³/s.




**Q1** yes — **Knows this step is needed?**
Tiago knows an order must be confirmed.

**Q2** yes — **Sees that the action is available?**
Apply is the only button in the panel.

**Q3** yes — **Connects the action to the effect?**
"Apply" means what he wants to do.

**Q4** no — **Sees progress after acting?**
The screen shows 120 at once, as if done. The gate takes about three minutes to move. No "ordered 120, current 92".

**Problem found (Q4).** Tiago cannot tell an accepted order from an executed one and may press Apply again. Severity 3, major.



11

# The record: one row per step (the table of Phase 1)

| Step | Correct action | Q1 | Q2 | Q3 | Q4 | Failure story: why Tiago would fail, and the evidence on the sketch | Sev. |
|---|---|---|---|---|---|---|---|
| 1 | Select Touvedo on the river map | yes | yes | yes | yes | Passes. The map is first on the screen, Touvedo is on it and gets highlighted. | 0 |
| 2 | Find the gate controls on the panel | yes | no | yes | yes | The setpoints look exactly like the read-only values around them. Nothing shows they can be edited. | 3 |
| 3 | Enter 120 in the flow field | yes | yes | no | yes | The field is labelled "Setpoint (%)". Tiago wants m³/s and does not know the opening-to-flow curve. | 4 |
| 4 | Check Ponte da Barca before confirming | no | yes | yes | yes | Nothing on the panel asks for a downstream check. The rule exists only in people's heads. | 3 |
| 5 | Confirm the order | yes | yes | yes | no | The screen shows 120 immediately, as if done. The gate takes three minutes. No ordered-vs-current display. | 3 |

**Severity scale (Nielsen, 1994). The same scale returns in Phase 2.**

- 0 not a usability problem
- 1 cosmetic
- 2 minor
- 3 major: important to fix
- 4 catastrophe: blocks the task

**A success story needs a reason too.** What on the sketch makes Tiago do the right thing? "He would just know" is not evidence!
If you cannot point at something on the sketch, the answer is no.

Hypothetical sketch of a COL gate panel. Nielsen, Severity ratings for usability problems, 1994



12

# Walk it through yourselves: the Andante machine

ACTIVITY · 5 min

**Persona:** a visitor on day one in Porto. **Task:** buy a ticket for one trip from Trindade to Campanhã.




**In pairs.** Ask the four questions at each screen. Which screen fails first, on which question, and why? One failure story per pair, written down.

**Then:** two pairs read their failure story aloud. Does the room agree on the screen and on the question number? Same sketch, different evaluators, different findings: that is why Phase 1 uses two or three evaluators.

Simplified screens. The real machine differs in detail, the zone step is the point



13

# The procedure: four roles and four rules

**Facilitator (design group)**
Shows the sketch of the step and says what is on it. Nothing else.

**Evaluators (2 or 3, another group)**
Take the persona's point of view and answer the four questions aloud, with the reason.

**Recorder (design group)**
Writes the table, one row per step. Every failure, including the obvious ones.

**Timekeeper**
Three minutes max. per step, then next.
(optional)

## Rules (Spencer, CHI 2000)

**No designing during the walkthrough.** Write the idea on a post-it and discuss it at the end.

**No defending the design.** The facilitator says what is on the sketch and nothing more.

**Write every failure.** Including the ones the designers find obvious.

**Three minutes per step.** If the argument runs longer, write it down as a problem and move on.

Without the rules the walkthrough turns into a design meeting, which is what Spencer saw happen at Microsoft.



14

# Phase 1 of the project: the walkthrough between groups

### Before
**the design group prepares**

- The persona: who the operator is and what they already know.
- Three missions from the briefing: one routine, one alarm, one with the coordinator.
- The correct action sequence of each mission, 5 to 10 steps, written down. Missing screens show up here.
- The sketches of every screen, in sequence order. One blank table per mission.

### During
**session S6, November**

- Group A walks through group B's sketches for about 15 minutes, then they swap.
- Facilitator and recorder from the design group, two or three evaluators from the other group.
- The four questions at every step. Three minutes per step. Spencer's rules.
- Ideas go on post-its, discussed only at the end.

### After
**what you hand over**

- The three filled tables, and half a page with the three worst problems, given to the design group the same day.
- The design group decides what changes before the wireframes of S7.
- Keep both: the tables you wrote for the other group, and the list of what you changed in your own sketches.

**Phase 2 (Module 10) adds the heuristics to the same screens.** The walkthrough asks "would this persona get through?" and the heuristic evaluation asks "does this screen break a known rule?". Sears (1997) combined the two as the heuristic walkthrough.



15

# Wrap-up

### A walkthrough is an inspection
someone **outside the design** simulates a first-time user, one step at a time, on paper, before any code exists

### Four questions at every step
right effect, action visible, action understood, progress seen. Spencer's two when time is short

### One case, five steps, four problems
a control that looks like a display, a percent where m³/s was expected, a rule that lives only in people's heads, a value shown as if done. One per question

### Before closing Phase 1
your group evaluates another group's sketches with this method. The tables and the changes are the evidence

**Next:** Module 07, visual representation of data.



16